Danny Yu

Full-Stack Engineer · Internal Platforms for STEM Education

danny_yu_2@sfu.ca · yannydu.github.io · github.com/yannydu

Summary

Full-stack engineer who builds and operates the internal platforms behind SFU's K-12 STEM outreach program: a publishing pipeline for camper projects, a curriculum system, and the lesson site and code playground used in class, running on Cloudflare Workers. Because the users are children, much of the engineering is privacy work: real names kept out of repos, databases, and backups by construction. Ships with Claude Code as a daily tool, owning the architecture, privacy model, and operations end to end.

Skills

Daily drivers	Python, JavaScript, HTML / CSS, Typst, git, Linux
Ship with (AI-assisted, still internalizing)	TypeScript, Astro / Hono, Cloudflare (Workers, D1, R2), Docker
AI-assisted development	Claude Code as a daily tool: custom skills, hooks, and subagents; taking a spec to a deployed system and owning what ships

Work Experience

SFU Science Alive (Faculty of Applied Sciences Outreach)
Burnaby, BC

School Programs and Instructional Lead
Jun. 2021 – Present

- Built **spark-gallery**, a publishing platform that turns each camp week into an issue of an online magazine, with camper-built games playable in the browser (Astro on Cloudflare Workers, D1, R2). Live through a full summer of camps: 509 camper projects from 416 campers, zero worker errors.
- Designed its privacy model for minors: camper names are pseudonymized before anything reaches the repo, and camper-written code is served from an isolated origin so it can never touch the gallery's cookies.
- Ship behind checks on every PR: typecheck, lint, `node --test` and Vitest suites (160+ test files), and a full-history gitleaks scan; a fail-closed pre-push name guard and EXIF/GPS stripping are tested invariants.
- Built **campkit**, the program's curriculum platform (Hono + Drizzle on Workers, D1, R2): 356 activities and 25 camps by six writers, written once, assembled into camps and costed into shopping lists.
- Grew **hello-wattson** (hellowattson.ca) from an internal lesson site into a code-gated teaching platform for outside teachers with zero student data: 22 programs, an in-browser Arduino compiler (Durable Object fronting a toolchain container), and **wattsons-playground**, a LiveCodes fork.
- Teach coding camps (Python, web dev, Arduino, LEGO robotics) and authored SPARK, the program's workshop-design framework.

Global Relay
Vancouver, BC

Software Development Engineer in Test (Co-op)
Jan. 2023 – Aug. 2023

- Built a Java JSON response builder for API validation and generated Java bindings from Thrift IDLs, integrating them with Wiremock containers for test stubs.
- Implemented a single source of truth for regression tests using the Trino REST API.

Incognito Software Systems Inc.
Vancouver, BC

QA Engineer Intern
Jan. 2022 – Aug. 2022

- Wrote automated GUI tests in Java and Selenium and set up a Docker + Jenkins continuous-testing environment.

Projects

ChessMate: A Smart Automated Chessboard
Burnaby, BC

SFU ENSC Capstone
Jan. 2024 – Aug. 2024

- Built the human-move detection module with a Raspberry Pi camera and OpenCV, comparing board states to identify the moved piece; drove an electromagnetic trolley in Python to execute the machine's moves.

Education

Simon Fraser University
Burnaby, BC

BASc in Computer Engineering
Graduated 2025